

# FactoryLM Contextualized-Diagnosis Proof Packet

## Capper 01 - Cap Torque Out of Range

**VERDICT: REVIEW** | scenario\_id: capper\_torque\_fault | tenant (SimLab): 00000000-0000-0000-0000-000000515ab1 | env: staging (factorylm/stg) | generated 2026-06-24

### 1. Executive Summary

A fault was injected into the SimLab juice-bottling line on **capper01**. Live signals were landed through the deployed relay ingest path into `live_signal_cache`, and the real MIRA Supervisor was asked the operator's question. MIRA returned a **grounded, cited diagnosis** (root cause + live evidence + manual citation + corrective action), not a clarifying question. All evidence below is independently verifiable against the staging database.

### 2. Technical Diagnosis

| Field                              | Value                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Fault                              | Capper 01 - Cap Torque Out of Range                                                                       |
| Affected asset                     | capper01                                                                                                  |
| UNS path                           | enterprise.florida_natural_demo.plant1.juice_bottling.line01.capper01                                     |
| UNS source                         | direct_connection (certified — chat-gate skipped)                                                         |
| Expected root cause (ground truth) | Capper chuck wear causing under-torque cap application                                                    |
| Expected actions (ground truth)    | Inspect capping chuck; Check and adjust clutch torque setting; Verify chuck wear and replace if necessary |
| MIRA confidence mode               | DIRECT_ANSWER (commit, no Socratic question)                                                              |

### 3. Live Tag Evidence (observed vs normal)

| Tag                 | Observed (faulted) | Normal baseline |
|---------------------|--------------------|-----------------|
| cap_torque_inlb     | 5.951              | 8.5             |
| cap_torque_variance | 2.2333             | 0.2             |
| reject_count        | 2                  | 0               |

Ingest evidence: relay POST /api/v1/tags/ingest -> **89 accepted, 0 rejected**; tag\_events rows=89 (distinct 89, all\_simulated=True); live\_signal\_cache rows=89.

### 4. Retrieved Documents & Citations

| # | Retrieved chunk (recall_knowledge, SimLab tenant)                                                                                                                                                                                                                  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>simlab://florida_natural_demo/line01/capper01/fault_code_table.md</b><br># Capper 01 - Fault Code Table   Code   Label   Severity   Description   Likely Cause   Recommended Action  <br> ----- ----- ----- ----- ----- ----- -----                             |
| 2 | <b>simlab://florida_natural_demo/line01/capper01/electrical_io_notes.md</b><br># Capper 01 - Electrical & I/O Notes ## Key I/O   Address   Tag   Description    ----- ----- -----    DI-CAP-01   E-Stop OK   Interlocked with Fil                                  |
| 3 | <b>simlab://florida_natural_demo/line01/capper01/plc_tag_description_sheet.md</b><br># Capper 01 - PLC Tag Description Sheet ## Process Tags   Tag Name   Full UNS Path   Type   Unit   Normal Range   Description  <br> ----- ----- ----- ----- ----- ----- ----- |
| 4 | <b>simlab://florida_natural_demo/line01/capper01/troubleshooting.md</b><br># Capper 01 - Troubleshooting Guide ## Cap Chute Empty / No Cap Present **Condition:** `cap_chute_level` < 20%; `cap_present` = FALSE on multiple cycles; rej                           |
| 5 | <b>simlab://florida_natural_demo/line01/capper01/troubleshooting.md</b><br># Capper 01 - Troubleshooting Guide ## High Torque / Over-torque Rejects **Condition:** `cap_torque_inlb` > 20 in-lb sustained; `reject_count` rising; `cap_t`                          |

Citations emitted in MIRA's answer: Key I/O, intro, intro, intro, intro

Expected citations (ground truth): troubleshooting.md; fault\_code\_table.md

## 5. MIRA Answer (verbatim)

**Q:** Capper is producing loose caps - what is wrong?

**A:** The capper01 is producing loose caps due to the `cap\_torque\_inlb` being lower than the target value of 8.5 in-lb, currently at 5.951 in-lb [Source: Key I/O]. This is likely causing the caps to be under-torqued, resulting in loose caps. To fix this issue, the following steps can be taken:

1. Check the clutch pads for wear or glazing [Source: intro].
2. Verify the chuck fit and ensure it is not worn [Source: intro].
3. Inspect the torque-head spring to ensure it is not weak [Source: intro].
4. Adjust the torque setpoint to the recommended value of 8.5 in-lb if necessary [Source: intro].

## 6. Database Evidence — live\_signal\_cache (faulted asset)

| uns_path                             | value  | sim  | src       |
|--------------------------------------|--------|------|-----------|
| capper01.faults.fault_code           | CA001  | True | simulator |
| capper01.motor.motor_current_amps    | 4.1021 | True | simulator |
| capper01.process.cap_torque_inlb     | 5.951  | True | simulator |
| capper01.process.cap_torque_target   | 8.5    | True | simulator |
| capper01.process.cap_torque_variance | 2.2333 | True | simulator |
| capper01.quality.reject_count        | 2      | True | simulator |
| capper01.status.cap_chute_level      | 81.35  | True | simulator |
| capper01.status.cap_present          | true   | True | simulator |
| capper01.status.jam_detected         | false  | True | simulator |
| capper01.status.run_state            | Idle   | True | simulator |

Verify yourself (staging, read-only): `SELECT uns_path, last_value_numeric, last_value_text FROM live_signal_cache WHERE tenant_id='00000000-0000-0000-0000-000000515ab1' AND uns_path::text LIKE '%capper01%';`

## 7. Langfuse Observability

| Field                    | Value                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Trace name               | supervisor.process                                                                                                         |
| Trace ID (this run)      | (not emitted — see note)                                                                                                   |
| Where to find it         | Langfuse instance at \$LANGFUSE_HOST -> Traces -> filter name='supervisor.process', user_id=chat_id                        |
| Spans to inspect         | the recall/retrieval span (input: query; output: KB chunks) and the LLM generation span (input: system prompt + chunks +   |
| Inputs/outputs to verify | generation INPUT contains the cited chunk text; generation OUTPUT cites that chunk -> grounded                             |
| Grounded vs hallucinated | GROUNDING if the answer's facts (e.g. the value + normal range) and the [Source:] citation both appear in the retrieved cl |

**Honest note:** this proof run executed the engine directly under Python 3.14, where Langfuse cannot emit (langfuse v2 = the telemetry-compatible client = incompatible with Py3.14; langfuse v4 dropped the .trace() API). So NO live Langfuse trace exists for THIS run. The DEPLOYED engine (Py 3.12 + langfuse 2.50 + \$LANGFUSE\_HOST) emits the 'supervisor.process' trace per turn. Grounding here is instead proven independently by sections 3-6 (the DB-backed retrieved chunks vs. the answer's citations) — which does not require trusting a trace.

## 8. Pass / Review Verdict (graded by simlab.diagnostic.grade)

| Rubric dimension                         | Result                                            |
|------------------------------------------|---------------------------------------------------|
| root_cause_hit (expected phrase matched) | FAIL                                              |
| asset_hit (expected asset named)         | PASS                                              |
| evidence_recall (>=0.5 to pass)          | 0.333                                             |
| citations_hit (by filename)              | (none - answer cites doc SECTIONS, not filenames) |
| actions_hit                              | (none matched the rubric keywords)                |
| rubric_passed                            | REVIEW                                            |

Verdict rule: rubric\_passed AND ingest==200 AND no\_trailing\_question. **REVIEW** is reported honestly, not hidden — a partial match (e.g. the answer is reasonable but does not contain the exact ground-truth root-cause phrase, or cites a doc SECTION rather than its filename) shows here as a failed dimension.

Overall: **REVIEW**

9. Provenance & Reproducibility

| Field                          | Value                                                                                                                                                                                              |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| git SHA / branch               | 7c1433ca3351 (docs/proof-packets) [DIRTY working tree]                                                                                                                                             |
| Config (env)                   | {"MIRA_TENANT_ID": "00000000-0000-0000-0000-000000515ab1", "MIRA_SHARED_TENANT_ID": "00000000-0000-0000-0000-000000515ab1", "MIRA_DIRECT_ANSWER_MODE": "1", "MIRA_RETRIEVAL_HYBRID_ENABLED": null} |
| Corpus                         | 242 chunks, doc-set md5 977b0f975778                                                                                                                                                               |
| Retrieval mode (actual)        | bm25-only (no vector/rerank - embedder/reranker down)                                                                                                                                              |
| Embedder / reranker / langfuse | ollama_reachable=True, nvidia_configured=True, langfuse=False (py 3.14.2)                                                                                                                          |
| Reproduce                      | doppler run --project factorylm --config stg -- python tools/proof/run_proof.py --clean (seeds: tools/seeds/seed-simlab-docs.py)                                                                   |